

Ayudantía 14- MAT1620

1. Use ordenadas polares para evaluar :

a) $\iint_D e^{-x^2-y^2} dA$ donde D es la región acotada por la semicircunferencia $x = \sqrt{4-y^2}$ y el eje Y .

b) $\int_0^1 \int_y^{\sqrt{2-y^2}} (x+y) dx dy$

2. Evalúe las siguientes integrales.

(a) $\int_0^{\sqrt{\pi}} \int_0^x \int_0^{xz} x^2 \sin(y) dy dz dx.$

(b) $\iiint_E e^{z/y} dV$ donde $E = \{(x, y, z) : 1 \leq y \leq 4, y \leq z \leq 4, 0 \leq x \leq z\}$

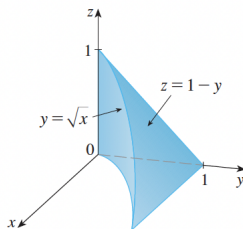
(c) $\iiint_E x dV$, donde E es el sólido que está acotado por el paraboloide $x = 4y^2 + 4z^2$ y el plano $x = 4$.

3. Use integrales triples para calcular el volumen del sólido encerrado por el cilindro $x^2 + y^2 = 4$ y los planos $y = -1$ y $z + y = 4$.

4. La figura muestra la región de integración para la integral

$$\int_0^1 \int_{\sqrt{x}}^1 \int_0^{1-y} f(x, y, z) dz dy dx$$

Reescriba en los otros cinco órdenes esta integral como una integral iterada equivalente.



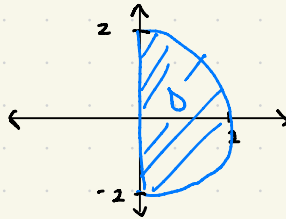
1. Use ordenadas polares para evaluar :

a) $\iint_D e^{-x^2-y^2} dA$ donde D es la región acotada por la semicircunferencia $x = \sqrt{4-y^2}$ y el eje Y .

b) $\int_0^1 \int_y^{\sqrt{2-y^2}} (x+y) dx dy$

$x^2 + y^2 = 4$

a) $\iint_D e^{-x^2-y^2} dA$



$$\begin{aligned} x &= r \cos(\theta) \\ y &= r \sin(\theta) \end{aligned} \quad \left\{ \begin{array}{l} 0 \leq r \leq 2, \\ -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2} \end{array} \right.$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^2 e^{-r^2} r dr d\theta$$

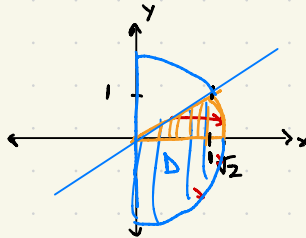
$u = r^2$
 $du = 2r dr \Rightarrow r dr = \frac{1}{2} du$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^2 \frac{1}{2} e^{-u} du d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left. -\frac{1}{2} e^{-u} \right|_0^2 d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(-\frac{1}{2} e^{-4} + \frac{1}{2} \right) d\theta$$

$$= \frac{\pi}{2} \left(-\frac{1}{2} e^{-4} + \frac{1}{2} \right) + \frac{\pi}{2} \left(-\frac{1}{2} e^{-4} + \frac{1}{2} \right) = \pi \left(-\frac{1}{2} e^{-4} + \frac{1}{2} \right) //$$

b) $\int_0^1 \int_y^{\sqrt{2-y^2}} (x+y) dx dy$

$x \leq \sqrt{2-y^2}$
 $x^2 + y^2 \leq 2$



$x = r \cos(\theta), y = r \sin(\theta)$

$0 \leq r \leq \sqrt{2}$

$0 \leq \theta \leq \frac{\pi}{4}$

$$\int_0^{\frac{\pi}{4}} \int_0^{\sqrt{2}} r (\cos(\theta) + \sin(\theta)) r \, dr \, d\theta$$

$$\int_0^{\frac{\pi}{4}} (\cos(\theta) + \sin(\theta)) \, d\theta \int_0^{\sqrt{2}} r^2 \, dr = \left(\sin(\theta) - \cos(\theta) \Big|_0^{\frac{\pi}{4}} \right) \cdot \frac{r^3}{3} \Big|_0^{\sqrt{2}}$$

$$= \left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} - 0 + 1 \right) \cdot \frac{2\sqrt{2}}{3} = \frac{2\sqrt{2}}{3} //$$

2. Evalúe las siguientes integrales.

$$(a) \int_0^{\sqrt{\pi}} \int_0^x \int_0^{xz} x^2 \sin(y) \, dy \, dz \, dx.$$

$$\begin{aligned} a) \int_0^{\sqrt{\pi}} \int_0^x \int_0^{xz} x^2 \sin(y) \, dy \, dz \, dx &= \int_0^{\sqrt{\pi}} \int_0^x -x^2 \cos(y) \Big|_0^{xz} \, dz \, dx \\ &= \int_0^{\sqrt{\pi}} \int_0^x -x^2 \cos(xz) + x^2 \, dz \, dx \\ &= \int_0^{\sqrt{\pi}} -\frac{x^2}{x} \sin(xz) + x^2 z \Big|_0^x \, dx = \int_0^{\sqrt{\pi}} -x \sin(x^2) + x^3 \, dx \\ &= - \int_0^{\sqrt{\pi}} x \sin(x^2) \, dx + \int_0^{\sqrt{\pi}} x^3 \, dx \end{aligned}$$

$$\hookrightarrow u = x^2$$

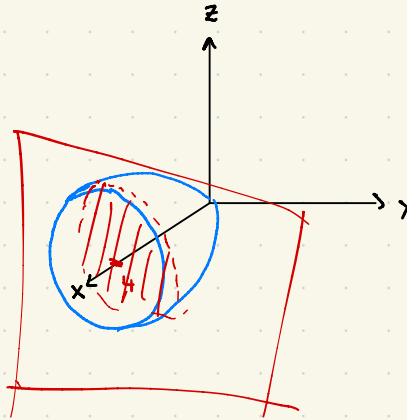
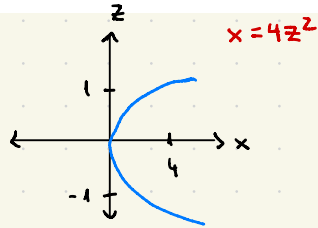
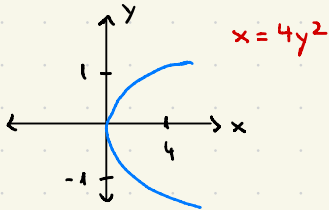
$$du = 2x \, dx \Rightarrow x \, dx = \frac{1}{2} du$$

$$= - \int_0^{\pi} \frac{1}{2} \sin(u) \, du + \frac{x^4}{4} \Big|_0^{\sqrt{\pi}} = \frac{1}{2} \cos(u) \Big|_0^{\pi} + \frac{\pi^2}{4} = \frac{\pi^2}{4} - 1 //$$

(c) $\iiint_E x \, dV$, donde E es el sólido que está acotado por el paraboloide $x = 4y^2 + 4z^2$ y el plano $x = 4$.

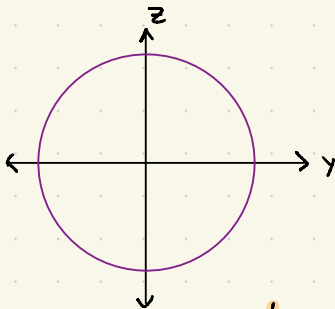
$$x = 4y^2 + 4z^2 \rightarrow \text{si } y = 0:$$

$$\hookrightarrow \text{si } z = 0:$$



$$\hookrightarrow \text{si } x = 4 \Rightarrow 4 = 4y^2 + 4z^2 \Rightarrow z^2 + y^2 = 1$$

$\hookrightarrow$ en el plano yz :



$$\hookrightarrow x = 4y^2 + 4z^2$$

$$\hookrightarrow 4z^2 = x - 4y^2$$

$$\hookrightarrow z^2 = \frac{x}{4} - y^2$$

$$\Rightarrow -\sqrt{\frac{x}{4} - y^2} \leq z \leq \sqrt{\frac{x}{4} - y^2}$$

$$x = 4y^2$$

$$\hookrightarrow \frac{x}{4} = y^2 \Rightarrow -\frac{\sqrt{x}}{2} \leq y \leq \frac{\sqrt{x}}{2}$$

$$0 \leq x \leq 4$$

$$\int_0^4 \int_{-\frac{\sqrt{x}}{2}}^{\frac{\sqrt{x}}{2}} \int_{-\sqrt{\frac{x}{4}-y^2}}^{\sqrt{\frac{x}{4}-y^2}} x \, dz \, dy \, dx$$

$$= \int_0^4 \int_{-\frac{\sqrt{x}}{2}}^{\frac{\sqrt{x}}{2}} x \sqrt{\frac{x}{4}-y^2} \, dy \, dx$$

$$\cos^2(\theta) + \sin^2(\theta) = 1$$

$$\Rightarrow \cos^2(\theta) = 1 - \sin^2(\theta)$$

$$\frac{x}{4} \cos^2(\theta) = \frac{x}{4} - \frac{x}{4} \sin^2(\theta)$$

$$\hookrightarrow y = \frac{\sqrt{x}}{2} \sin(\theta) \quad \hookrightarrow dy = \frac{\sqrt{x}}{2} \cdot \cos(\theta) \, d\theta$$

$$\hookrightarrow -\frac{\sqrt{x}}{2} \leq y \leq \frac{\sqrt{x}}{2}$$

$$-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

$$= \int_0^4 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2x \frac{\sqrt{x}}{2} \cos(\theta) \cdot \frac{\sqrt{x}}{2} \cos(\theta) \, d\theta \, dx$$

$$= \int_0^4 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^2}{2} \cos^2(\theta) \, d\theta \, dx = \int_0^4 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^2}{2} \left(\frac{1 + \cos(2\theta)}{2} \right) \, d\theta \, dx$$

$$= \int_0^4 \frac{x^2}{2} \, dx \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2} + \frac{1}{2} \cos(2\theta) \, d\theta = \frac{x^3}{6} \Big|_0^4 \cdot \left(\frac{1}{2} \theta + \frac{1}{4} \sin(2\theta) \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \right)$$

$$= \frac{32}{3} \left(\frac{\pi}{4} + \frac{\pi}{4} \right) = \frac{16\pi}{3}$$